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AUTO-ASSEMBLED-AND-FOLDED SUPPORT FRAME

Abstract

A support frame that can be assembled automatically comprises multiple straight rod assemblies and multiple linkage assemblies. The straight rod assemblies are arranged at intervals in a rectangular shape. Each linkage assemblies connects two straight rod assemblies and comprises multiple upper rods, one joint seat connected to the upper rods, multiple lower rods, and one connecting seat connected to the lower rods. When the joint seat is attached to the connecting seat, the support frame and the straight rod assemblies are expanded. When the joint seat is detached from the connecting seat, the support frame and the straight rod assemblies are folded. All aforementioned actions can be easily achieved by operating a pressure ring assembly, which is already attached on the straight rod assemblies. The support frame is convenient for storage, saves transportation space, and can be automatically stacked and set up scaffolding with automated equipment.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a support frame, especially to a supporting frame that is used in construction and can be automatically assembled and folded.

2. Description of Related Art

[0002] In construction nowadays, for high-floor constructing and building, scaffolding is assembled for workers to climb up and down. Most of the scaffolding of lower-level floors, such as floors under the third floor, is assembled manually. Lower-level floors scaffolding is assembled firstly by composing a frame, which is made by assembling multiple rods to four posts, then mounting platforms onto the frame, and so on to build to a preset height. While assembling the scaffolding, workers must transfer components of the scaffolding upward to assemble, then get down to retrieve more components, repeating the aforementioned movements again and again until the scaffolding building is finished.

[0003] However, in proceeding of stacking the scaffolding, the workers have to climb up and down, thus consuming lots of energy and time. Furthermore, climbing while carrying things is dangerous, especially when assembling the frame at higher levels and without security fences. Besides, for convenience, the workers may neglect wearing safety aids when the highness they are at is not so high, causing larger risk of work accident. Therefore, there are needs of improvement.

SUMMARY OF THE INVENTION

[0004] In view of shortcomings of the prior arts, the present invention provides an auto-assembled-and-folded support frame that can be assembled and folded automatically. The support frame can be quickly expanded or folded by its multiple straight rod assemblies, and multiple linkage assemblies are connected to be expandable and foldable. The present invention can increase safety for assembling and dismantling of scaffolding.

[0005] The support frame comprises the multiple straight rod assemblies and the multiple linkage assemblies. The multiple straight rod assemblies are arranged at intervals in a rectangular shape. Each one of the multiple linkage assemblies is connected between two of the multiple straight rod assemblies and comprises at least one first upper rod, a second upper rod, a joint seat, a first lower rod, at least one second lower rod, and a connecting seat. The at least one first upper rod and the second upper rod each have a bottom end pivotally connected to the joint seat. A top end of the at least one first upper rod is pivotally connected to one of the two corresponding straight rod assemblies. A top end of the second upper rod is pivotally connected to the other one of the two corresponding straight rod assemblies. The first lower rod and the at least one second lower rod each have a top end pivotally connected to the connecting seat. A bottom end of the first lower rod pivotally is connected to one of the two corresponding straight rod assemblies. A bottom end of the at least one second lower rod is pivotally connected to the other one of the two corresponding straight rod assemblies. The connecting seat is disposed on top of the joint seat. The connecting seat and the joint seat are attached to or detached from each other selectively. When the support frame is in an expanded state, the joint seat and the connecting seat of each one of the multiple linkage assemblies are attached to each other. When the support frame is in a folded state, the joint seat and the connecting seat of each one of the multiple linkage assemblies are detached from each other. Distances between the multiple straight rod assemblies when the support frame is in the expanded state is longer than distances between the multiple straight rod assemblies when the support frame is in the folded state.

[0006] Furthermore, the support frame comprises a pressure ring assembly comprising a ring body and multiple support rods. Each one of the multiple support rods has an inner end pivotably connected to the ring body and an outer end pivotably connected to one of the multiple straight rod assemblies. A connecting position of said support rod and said straight rod assembly is disposed between two connecting positions. Among them, one of the two connecting positions is between the at least one first upper rod and said straight rod assembly, and the other one of the two connecting positions is between the first lower rod and said straight rod assembly. Or, one of the two connecting positions is between the second upper rod and said straight rod assembly, and the other one of the two connecting positions is between the at least one second lower rod and said straight rod assembly.

[0007] Advantages of the support frame are that the support frame has the expanded state and the folded state. Therefore, the support frame can be folded for easy transportation and saving space, and can be expanded quickly for stacking scaffolding. Further, by moving the ring body, the support frame can be expanded simply and fast for stacking scaffolding. Besides, when stacking scaffolding with multiple support frames, the support frames can be automatically assembled by using a base rack set and at least one robotic arm, thereby significantly reducing labor costs and building time, hence increasing work safety.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a support frame that can be assembled and folded automatically in accordance with the present invention in an expanded state;

[0009] FIG. 2 is a partially-enlarged exploded view of a straight rod assembly of the support frame in accordance with the present invention;

[0010] FIG. 3 is a partially-enlarged perspective view of the support frame in accordance with the present invention;

[0011] FIG. 4 is a partially exploded view of elements related to a joint seat and a connecting seat of the support frame in accordance with the present invention;

[0012] FIG. 5 is a partial sectional view of the joint seat and the connecting seat, being detachable, of the support frame in accordance with the present invention;

[0013] FIG. 6 is a partial sectional view of the joint seat and the connecting seat, being attached and in a locked-in-place state, of the support frame in accordance with the present invention;

[0014] FIG. 7 is a partial sectional view of the joint seat and the connecting seat of the support frame in accordance with the present invention, when a plug does not engage;

[0015] FIG. 8 is a partial sectional view of the joint seat and the connecting seat of the support frame in accordance with the present invention, when the plug is engaged;

[0016] FIG. 9 is a partially-enlarged exploded view of a pressure ring assembly of the support frame in accordance with the present invention;

[0017] FIG. 10 is a partially-enlarged exploded view of a support rod of the support frame in accordance with the present invention, mounted to the straight rod assembly;

[0018] FIG. 11 is a side view of the support frame in accordance with the present invention, shown expanded;

[0019] FIG. 12 is a side view of the support frame in accordance with the present invention, shown folded;

[0020] FIG. 13 is a perspective view of the support frame in accordance with the present invention, used to build scaffolding; and

[0021] FIG. 14 is a partially-enlarged view of FIG. 13.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0022] With reference to drawings and preferable embodiments in accordance to the present invention, descriptions below are to elaborate technical means to achieve intended purpose of the present invention.

[0023] With reference to FIG. 1, a support frame that can be assembled and folded automatically in accordance with the present invention comprises multiple straight rod assemblies **10**, multiple linkage assemblies **2** and a pressure ring assembly **90**.

[0024] With reference to FIGS. 1, 2, 11 and 14, in this embodiment, the support frame has four straight rod assemblies **10** arranged at intervals in a rectangular shape. One of the multiple linkage assemblies **2** is disposed between two of the four straight rod assemblies **10**, which are disposed at a same side of the support frame. Each one of the multiple linkage assemblies **2** comprises at least one first upper rod **40**, a second upper rod **41**, a joint seat **20**, a first lower rod **50**, at least one second lower rod **51**, and a connecting seat **30**. In this embodiment, each one of the multiple straight rod assemblies **10** comprises a vertical post **11**, multiple upper contact seats **12**, multiple middle contact seats **13**, and multiple lower contact seats **14**. The vertical post **11** has a support disc **15** mounted at a bottom end of the vertical post **11**. The support disc **15** has multiple engagement holes **151** penetrating through the support disc **15**, arranged around a center of the support disc **15**, and spaced apart from each other. But it does not limit to that. Each one of the multiple upper contact seats **12** is a plate and has at least one upper pivotally-connecting hole **121** penetrating through said upper contact seat **12**. One end of each one of the multiple upper contact seats **12** is mounted at the vertical post **11** and is near a top end of the vertical post **11**. The multiple upper contact seats **12** are arranged around an axial center of the vertical post **11** and are spaced apart from each other. Each one of the multiple middle contact seats **13** is a plate and has at least one middle fixation hole **131** penetrating through said middle contact seat **13**. The multiple middle contact seats **13** are arranged around the axial center of the vertical post **11** and are spaced apart from each other. Each one of the multiple lower contact seats **14** is a plate and has at least one lower pivotally-connecting hole **141** penetrating through said lower contact seat **14**. The multiple lower contact seats **14** are arranged around the axial center of the vertical post **11** and are spaced apart from each other. However, configurations of the multiple straight rod assemblies **10** are not limited to that and can be altered according to user's needs.

[0025] With reference to FIGS. 1, 3 and 4, for conciseness in description, the following descriptions take two of the multiple straight rod assemblies **10** and one of the multiple linkage assemblies **2** that are disposed at the same side of the support frame as an example. The joint seat **20** comprises an upper pivotally-connecting portion **21** and a tenon **22**. The upper pivotally-connecting portion **21** is columnar and has a front-side surface, a rear-side surface, and two lateral-side surfaces, which are disposed between the front-side surface and the rear-side surface. The upper pivotally-connecting portion **21** has two upper pivotally-connecting pieces **211** respectively and laterally protruding from the front-side surface and the rear-side surface of the upper pivotally-connecting portion **21**. Each one of the two upper pivotally-connecting pieces **211** has an upper contact hole **212** penetrating through said upper pivotally-connecting piece **211**. The tenon **22** protrudes from an end of the upper pivotally-connecting portion **21** and has a clip recess **221** surrounding an outer wall of the tenon **22** and disposed at an end of the tenon **22** that is near the upper pivotally-connecting portion **21**. The tenon **22** has at least one alignment recess **222** depressed to form on the outer wall of the tenon **22** and extending vertically. In this embodiment, an amount of the at least one alignment recess **222** is multiple, and the multiple alignment recesses **222** extend coaxially along the upper pivotally-connecting portion **21** and are arranged opposite to each other.

[0026] With reference to FIGS. 1, 3 and 4 to 8, the connecting seat **30** comprises a lower pivotally-connecting portion **31** and a lock plate **32**. The lower pivotally-connecting portion **31** is columnar and has a contact recess **311** depressed to form at an end of the lower pivotally-connecting portion **31**. A shape of the contact recess **311** corresponds to a shape of the tenon **22**. The lower pivotally-

connecting portion **31** has a front-side surface, a rear-side surface, and two lateral-side surfaces, which are disposed between the front-side surface and the rear-side surface of the lower pivotally-connecting portion **31**. The lower pivotally-connecting portion **31** has two lower pivotally-connecting pieces **312** respectively and laterally protruding from the front-side surface and the rear-side surface of the lower pivotally-connecting portion **31**. Each one of the two lower pivotally-connecting pieces **312** has a lower contact hole **313** penetrating through said lower pivotally-connecting piece **312**. The connecting seat **30** has a guide recess **314** depressed to form between the two lower pivotally-connecting pieces **312** and extending vertically on the connecting seat **30** and coaxially along the lower pivotally-connecting portion **31**. But the connecting seat **30** is not limited to that and may have no guide recess **314**. The lock plate **32** has a limiting hole **321** penetrating through it. The lock plate **32** has at least one protrusion rib **322** protruding to form at a contour of the limiting hole **321**. In this embodiment, the lock plate **32** has two protrusion ribs **322** disposed at two opposite sides of the contour of the limiting hole **321** and corresponding to the alignment recess **222** in position. The lock plate **32** has an elastic element **323** mounted at a side of the contour of the limiting hole **321**. In this embodiment, the elastic element **323** is a compression spring. At another side, which is opposite to the elastic element **323**, of the contour of the limiting hole **321**, a slot **324** is depressed to form thereon and communicates with the guide recess **314**. But it is not limited to that, as configurations of the elastic element **323** and whether to have the slot **324** can be altered according to user's needs. The lock plate **32** is mounted at the lower pivotally-connecting portion **31** and is near an opening of the contact recess **311**. The tenon **22** can pass through the limiting hole **321** to reach into the contact recess **311**. Pushed by the elastic element **323**, the tenon **22** can be moved along the limiting hole **321** relative to the connecting seat **30** until an end of the contour of the limiting hole **321** contacts the clip recess **221** of the tenon **22**. The joint seat **20** is disposed below the connecting seat **30**, but it is not limited to that, and structures of the joint seat **20** and structures of the connecting seat **30** can be exchanged, which means the tenon **22** can be formed at the connecting seat **30** and the contact recess **311** can be formed at the joint seat **20**.

[0027] In this embodiment, an amount of the at least one first upper rod **40** is two, an amount of the at least one second lower rod **51** is two, and said rods are inclined. The two first upper rods **40** are parallel and the two second lower rods **51** are parallel. Bottom ends of the two first upper rods **40** and a bottom end of the second upper rod **41** are pivotably mounted at the joint seat **20** through pins **60**, which are pivotably mounted at the upper contact holes **212** of the two upper pivotally-connecting pieces **211** of the joint seat **20**. Top ends of the two first upper rods **40** are pivotably mounted at one of the two straight rod assemblies **10**, which are disposed at the same side of the support frame, and specifically mounted at the upper pivotally-connecting hole **121** of the upper contact seat **12**. A top end of the second upper rod **41** is pivotably mounted at the other one of the two straight rod assemblies **10**, and specifically mounted at the upper pivotally-connecting hole **121** of the upper contact seat **12**. A top end of the first lower rod **50** and top ends of the two second lower rods **51** are pivotably mounted at the connecting seat **30**, specifically are pivotably mounted at the lower contact holes **313** of the two lower pivotally-connecting pieces **312** of the connecting seat **30** through other pins **60**. A bottom end of the first lower rod **50** is pivotably mounted at one, which is the same one that the two first upper rods **40** are connected to, of the two straight rod assemblies **10**, and specifically is pivotably mounted at the lower pivotally-connecting hole **141** of the lower contact seat **14**. Bottom ends of the two second lower rods **51** are pivotably mounted at the other one of the two straight rod assemblies **10**, and specifically are pivotably mounted at the lower pivotally-connecting hole **141** of the lower contact seat **14**. The first lower rod **50** is disposed between the two first upper rods **40**. The second upper rod **41** is disposed between the two second lower rods **51**. By this two-rods configuration, structural stability is enhanced to avoid torsion. But it is not limited to that.

[0028] With reference to FIGS. **1**, **9** and **10**, the pressure ring assembly **90** comprises a ring body

91 and multiple support rods **92**. The ring body **91** has an outer surrounding surface and multiple connecting portions **93** protruding from the outer surrounding surface. In this embodiment, an amount of the multiple connecting portions **93** is four, but it is not limited to that. Each one of the multiple connecting portions **93** has a through hole **94** penetrating through it. An inner end of each one of the multiple support rods **92** is pivotably mounted at one of the multiple connecting portions **93** by a first pivotable bolt **95** passing through the through hole **94** of said connecting portion and said support rod **92**. An outer end of each one of the multiple support rods **92** is pivotably mounted at the middle contact seat **13** of one of the multiple straight rod assemblies **10** through a second pivotable bolt **96**. Pivotable connecting is common technique, and therefore is not to be further elaborated.

[0029] Each one of the multiple straight rod assemblies **10** has an upper portion, a lower portion and a supporting portion. The upper portion (upper contact seat **12**) is connected to the at least one first upper rod **40** or the second upper rod **41**. The lower portion (lower contact seat **14**) is connected to the first lower rod **50** or the at least one second lower rod **51**. The supporting portion (middle contact seat **13**) is connected to one of the support rods **92**, and the supporting portion is disposed between the upper portion and the lower portion.

[0030] The support frame can be in an expanded state or a folded state. With reference to FIGS. **1** and **11**, when the support frame is in the expanded state, any two of the multiple straight rod assemblies **10** keep a distance between them. The at least one first upper rod **40** and the second lower rod **51** disposed at the same side of the support frame are parallel. The at least one first lower rod **50** and the second upper rod **41** disposed at the same side of the support frame are parallel. Therefore, the upper rods **40**, **41** and the lower rods **50**, **51** compose an X-shaped structure. The ring body **91** of the pressure ring assembly **90** is approximately disposed at a central position of the support frame, and the multiple support rods **92** are horizontal, so the pressure ring assembly **90** can stably support and unfold the multiple straight rod assemblies **10**, enhancing structural rigidity. With reference to FIGS. **4**, **6** and **7**, the tenon **22** passes through the limiting hole **321** and reaches into the contact recess **311**. By the elastic element **323** pushing the tenon **22**, the clip recess **221** of the tenon **22** is moved in the lock plate **32** until contacting an end of the limiting hole **321** to engage with the lock plate **32**. By the clip recess **221** engaged with the lock plate **32** to fix it in place, the joint seat **20** and the connecting seat **30** are attached and fixed together to be in a locked-in-place state.

[0031] With reference to FIGS. **4**, **5** and **8**, the support frame can change its configuration from the expanded state to the folded state through using a plug **70**. The plug **70** has a vertical edge **71** and an inclined edge **72** disposed opposite to each other. The inclined edge **72** contacts the lock plate **32**. The plug **70** is inserted into the slot **324** along the guide recess **314** to move the joint seat **20** toward the elastic element **323**, so the joint seat **20** compresses the elastic element **323** until the at least one protrusion rib **322** aligns with the at least one alignment recess **222**. Therefore, the joint seat **20** and the connecting seat **30** are detachable, and can be respectively moved away from each other vertically to be detached. In other words, the plug **70** works as a security mechanism. Only when the user is operating the plug **70** deliberately, can the support frame be folded. Without operating the plug **70**, the support frame maintains the expanded state and cannot be folded. Distances between the multiple straight rod assemblies **10** when the support frame is in the expanded state is longer than distances between the multiple straight rod assemblies **10** when the support frame is in the folded state. With reference to FIG. **12**, while the joint seat **20** and the connecting seat **30** are moved away from each other and in process of detachment, the at least one first upper rod **40**, the second upper rod **41**, the first lower rod **50**, the at least one second lower rod **51**, and the multiple support rods **92** are pivoted at the same time, so the multiple straight rod assemblies **10** are all moved toward each other. The ring body **91** is moved downward until the multiple straight rod assemblies **10** and the multiple support rods **92** are near each other and the support frame is at the folded state. Therefore, the support frame is folded for easy transportation

and saving space. By moving the ring body **91**, the support frame and the multiple straight rod assemblies **10** can be expanded quickly for stacking scaffolding. To change the support frame from the folded state to the expanded state, simply push the ring body **91** upward, and the multiple support rods **92** will simultaneously push the multiple straight rod assemblies **10** outwardly. To change the support frame from the expanded state to the folded state, simply pushing the ring body **91** downward can achieve it.

[0032] With reference to FIGS. **11**, **13** and **14**, when stacking scaffolding with multiple support frames, a base rack set **80** and at least one robotic arm can be used for stacking with appropriate human assistance. The base rack set **80** comprises a base surrounding frame **81** and multiple extendable racks **82**. The **22** surrounding frame is rectangular. Each one of the multiple extendable racks **82** mounted at one of four corners of the base surrounding frame **81** can extend upwardly, and has multiple segments. The segment of the extendable rack **82** has a support portion **821** near a bottom end of the segment and having multiple engaging elements that engage with the engagement holes **151** of the support disc **15** of the vertical post **11**. Structural details will not be further elaborated since they are common technique. The robotic arm is mounted near the base rack set **80**. The robotic arm expands the first one of the multiple support frames from the folded state to the expanded state and puts the first support frame on the base rack set **80**. Each one of the multiple straight rod assemblies **10** is disposed inside one of the multiple extendable racks **82**. The multiple engagement holes **151** of the support disc **15** are engaged with the support portion **821**. Multiple horizontal racks are mounted on the lower contact seats **14** of different straight rod assemblies **10** and are stacked on the upper contact seats **12** of different straight rod assemblies **10**. Multiple platform boards are putted on the multiple horizontal racks. According to user's needs, the horizontal racks can be mounted on the at least one middle fixation hole **131** of the middle contact seats **13** of two of the multiple straight rod assemblies **10** that are at the same side of said support frame. When completing assembling the first support frame, the first segments of the multiple extendable racks **82** are lifted simultaneously. Then the robotic arm expands and puts the second support frame between the second segments of the multiple extendable racks **82** to be assembled with aforementioned process, and so on. The multiple straight rod assemblies **10** of one of the support frames are respectively engaged with the multiple straight rod assemblies **10** of another one of the support frames that is on top of them, which is common technique and hence will not be further elaborated. Therefore, the support frames can be automatically assembled. During building construction, the robotic arm and operators can complete assembling on the ground without climbing up and down to proceed stacking, thereby significantly reducing risks and improving work safety. Besides, minimal manpower and fast building of scaffolding can reduce labor costs.

[0033] Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the invention, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. An auto-assembled-and-folded support frame comprising: multiple straight rod assemblies arranged at intervals in a rectangular shape; multiple linkage assemblies, each one of the multiple linkage assemblies connected between two of the multiple straight rod assemblies and comprising at least one first upper rod; a second upper rod; a joint seat; a first lower rod; at least one second lower rod; a connecting seat; the at least one first upper rod and the second upper rod each having a bottom end pivotally connected to the joint seat; the at least one first upper rod having a top end pivotally connected to one of the two corresponding straight rod assemblies; the second upper rod

having a top end pivotally connected to the other one of the two corresponding straight rod assemblies; the first lower rod and the at least one second lower rod each having a top end pivotally connected to the connecting seat; the first lower rod having a bottom end pivotally connected to one of the two corresponding straight rod assemblies; the at least one second lower rod having a bottom end pivotally connected to the other one of the two corresponding straight rod assemblies; the connecting seat disposed on top of the joint seat, and the connecting seat and the joint seat attached to or detached from each other selectively; wherein when the auto-assembled-and-folded support frame is in an expanded state, the joint seat and the connecting seat of each one of the multiple linkage assemblies are attached to each other; wherein when the auto-assembled-and-folded support frame is in a folded state, the joint seat and the connecting seat of each one of the multiple linkage assemblies are detached from each other; wherein distances between the multiple straight rod assemblies when the auto-assembled-and-folded support frame is in the expanded state is longer than distances between the multiple straight rod assemblies when the auto-assembled-and-folded support frame is in the folded state.

2. The auto-assembled-and-folded support frame as claimed in claim 1, wherein in each one of the multiple linkage assemblies, an amount of the at least one second lower rod of is two, and the second upper rod is disposed between the two second lower rods.

3. The auto-assembled-and-folded support frame as claimed in claim 2, wherein in each one of the multiple linkage assemblies, the connecting seat comprises a lower pivotally-connecting portion having a front surface; a rear surface; the two second lower rods respectively and pivotally mounted at the front surface and the rear surface of the lower pivotally-connecting portion; and two lateral surfaces disposed between the front surface and the rear surface.

4. The auto-assembled-and-folded support frame as claimed in claim 1, wherein each one of the multiple linkage assemblies further comprises an engagement mechanism mounted between the connecting seat and the joint seat; a plug detachably inserted into the engagement mechanism; and when the plug is detached from the engagement mechanism, the engagement mechanism stops the connecting seat and the joint seat, which are attached, from being detached vertically from each other; when the plug is inserted into the engagement mechanism, the engagement mechanism allows the connecting seat and the joint seat, which are attached, to be detached vertically from each other.

5. The auto-assembled-and-folded support frame as claimed in claim 4, wherein the connecting seat comprises a contact recess; the joint seat comprises a tenon detachably located in the contact recess; when the connecting seat and the joint seat are attached to each other, the tenon reaching into the contact recess.

6. The auto-assembled-and-folded support frame as claimed in claim 5, wherein the tenon has a clip recess surrounding an outer wall of the tenon; the engagement mechanism further comprises a lock plate mounted at a bottom end of the joint seat and having a limiting hole penetrating through the lock plate; when the joint seat and the connecting seat are attached to each other, the tenon passing through the limiting hole, and a position of a contour of the limiting hole corresponding to a position of the clip recess; an elastic element mounted between the lock plate and the tenon, and pushing to move the lock plate and the tenon relatively, such that the contour of the limiting hole of the lock plate reaches into the clip recess to prevent the tenon from being detached from the connecting seat; when the plug is inserted into the connecting seat, the plug pushing to move the lock plate and the tenon relatively, such that the contour of the limiting hole of the lock plate is detached from the clip recess to allow the tenon to be detached from the connecting seat.

7. The auto-assembled-and-folded support frame as claimed in claim 6, wherein the connecting seat has a guide recess extending vertically on the connecting seat; the lock plate has a slot depressed to form on the contour of the limiting hole of the lock plate and communicating with the guide recess; the plug is detachably mounted through the slot and the guide recess to push the tenon to move relative to the lock plate.

8. The auto-assembled-and-folded support frame as claimed in claim 6, wherein the tenon has at least one alignment recess depressed to form on the outer wall of the tenon and extending vertically; the lock plate has at least one protrusion rib protruding to form at the contour of the limiting hole; after the plug is inserted into the connecting seat to push to move the lock plate and the tenon relatively, a position of the at least one protrusion rib corresponding to a position of the at least one alignment recess.

9. The auto-assembled-and-folded support frame as claimed in claim 8, wherein the plug has a vertical edge and an inclined edge disposed opposite to each other; the inclined edge contacting the lock plate.

10. The auto-assembled-and-folded support frame as claimed in claim 1, wherein each one of the multiple straight rod assemblies comprises a vertical post having a support disc mounted at a bottom end of the vertical post; multiple upper contact seats; one end of each one of the multiple upper contact seats mounted at the vertical post; each one of the multiple upper contact seats pivotably connected to the top end of the corresponding at least one first upper rod or connected to the top end of the corresponding second upper rod; and multiple lower contact seats; one end of each one of the multiple lower contact seats mounted at the vertical post; each one of the multiple lower contact seats pivotably connected to the bottom end of the corresponding first lower rod or connected to the bottom end of the corresponding at least one second lower rod.

11. The auto-assembled-and-folded support frame as claimed in claim 9, wherein each one of the multiple straight rod assemblies comprises a vertical post having a support disc mounted at a bottom end of the vertical post; multiple upper contact seats; one end of each one of the multiple upper contact seats mounted at the vertical post; each one of the multiple upper contact seats pivotably connected to the top end of the corresponding at least one first upper rod or connected to the top end of the corresponding second upper rod; and multiple lower contact seats; one end of each one of the multiple lower contact seats mounted at the vertical post; each one of the multiple lower contact seats pivotably connected to the bottom end of the corresponding first lower rod or connected to the bottom end of the corresponding at least one second lower rod.

12. The auto-assembled-and-folded support frame as claimed in claim 1 further comprising: a pressure ring assembly comprising a ring body; multiple support rods; each one of the multiple support rods having an inner end pivotably connected to the ring body; an outer end pivotably connected to one of the multiple straight rod assemblies; wherein each one of the multiple straight rod assemblies has an upper portion connected to the at least one first upper rod or the second upper rod; a lower portion connected to the first lower rod or the at least one second lower rod; a supporting portion connected to one of the support rods, and disposed between the upper portion and the lower portion.
